Ni/Zeolite-Y 촉매 기반 Polypropylene 고압 열분해 수소 생산

Hydrogen Production via High-Pressure Pyrolysis of Polypropylene Using Ni-Supported Zeolite Y Catalyst

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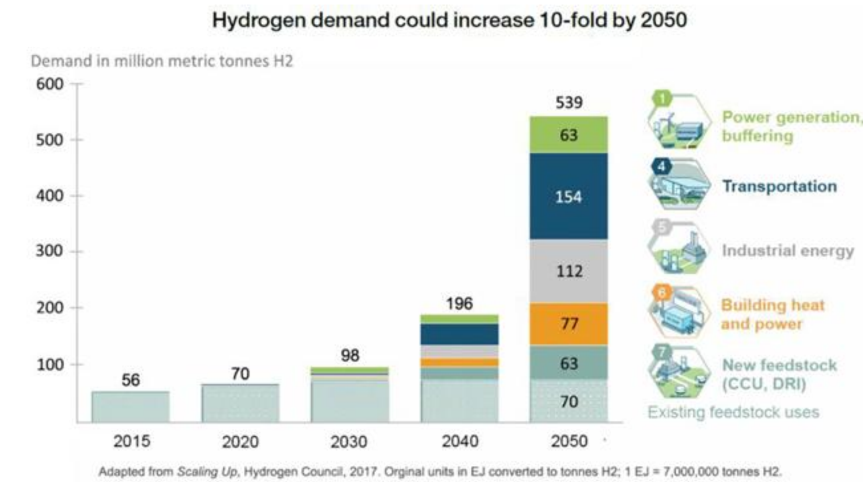
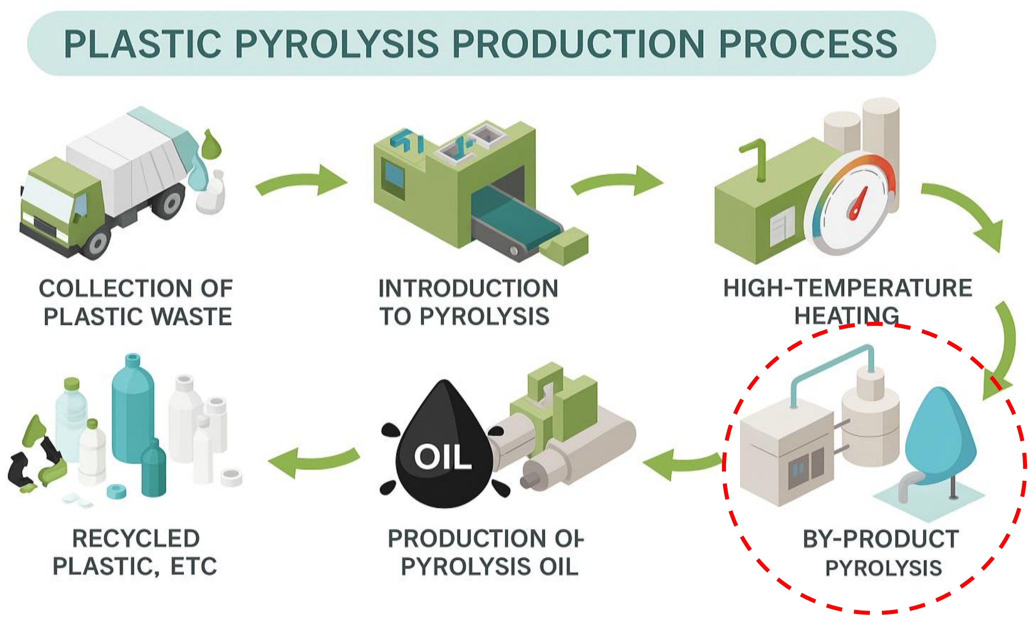
Contents

Research background





Literature review

PCERI

Characteristics of plastic and catalyst

Ref. A review on catalytic pyrolysis of plastic wastes to high-value products., P.Yujie, *Energy Conversion Management*, 2022

Product distributions in the pyrolysis of different plastics.

Reference	Feedstock	Catalyst	Condition*	Yields (wt.%) Gas	Yields (wt.%) Oil
[37]	HDPE	None	T = 650 °C Time = 0.8 s	20.3	–
			T = 650 °C Time = 1.5 s	31.5	–
[23]	PP	None	T = 746 °C	65.9	29.6
[23]	PE	None	T = 728 °C	59.3	38.2

– Selection of **polypropylene** through comparative analysis

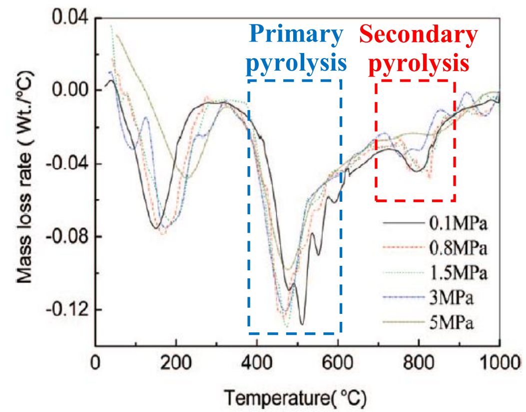
Product distributions obtained by different authors in the pyrolysis with different catalysts.

Reference	Feedstock	Catalyst	Condition	Yields (wt.%) Gas	Yields (wt.%) Oil
[74]	LDPE	ZSM-5	T_1 = 480 °C T_2 = 375 °C	72.73	24.94
[75]	HDPE	Y zeolite	T = 450 °C Time = 30 min	71.5	27
[70]	HDPE	HZSM-5 Hβ HY	T = 500 °C Time = 15 h	57 36 –	41.94 30.8 –
[81]	PP	HZSM-5 HUSY HMOR	T = 360 °C	94.77 89.49 88.29	2.31 3.75 4.54
[84]	HDPE	SBA-16 Al-SBA-16	T = 430 °C Time = 2 h	6.3 20.6	43.3 75.8

– Verification of increased gas emission at lower temperatures due to catalytic effects

Primary pyrolysis and secondary pyrolysis

Ref. Influence of pressure on coal pyrolysis and char gasification, H. Yang, *Energy&Fuels*, 2007



DTG curves of pyrolysis under different pressures

- Primary Pyrolysis (400-700°C)

– Production of CO, CO_2, CH_4, H_2O through organic reactions

– **Inhibition** of volatile matter release due to high pressure conditions

- Secondary Pyrolysis (over 700°C)

– Production of H_2 and CO through internal tar cracking

– Increased residence time of volatile matter due to high pressure conditions

– **Enhanced** hydrocarbon cracking and gas production promotion

H_2 production with Ni/zeolite-Y catalyst

Ref. Co-production of hydrogen and carbon nanofibers …, M,d.Nasir Uddin, *Energy Conversion and Management*, 2015

Results from methane decomposition over Ni/Zeolite-Y

No.	T(°C)	Initial H_2 (%)	Initial CH_4 (%)	No.	T(°C)	Initial H_2 (%)	Initial CH_4 (%)
15% Ni-supported Y zeolite catalyst				30% Ni-supported Y zeolite catalyst			
1	**500**	**19.09**	**10.52**	1	**500**	**18.08**	**9.91**
2	550	30.27	17.73	2	550	34.85	20.82
3	600	36.32	22.11	3	600	48.85	31.57
4	**650**	**41.02**	**25.76**	4	**650**	**58.58**	**41.67**

Behavior of AC (Accumulated carbon) with total methane conversion

No.	T(°C)	Total methane conversion(mol/g_{cat})	No.	T(°C)	Total methane conversion(mol/g_{cat})
15% Ni-supported Y zeolite catalyst			15% Ni-supported Y zeolite catalyst		
1	500	197.08	1	500	482.56
2	550	526.40	2	550	1047.30
3	**600**	**586.21**	3	**600**	**1242.54**
4	**650**	**405.47**	4	**650**	**821.67**

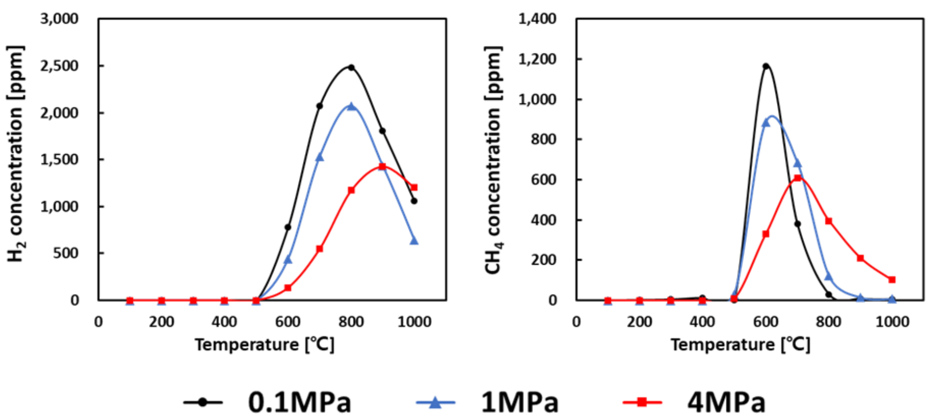
– **Increased H_2** and **CH_4** production with temperature rise

– H_2 with 15% Ni/zeolite-Y < H_2 with 30% Ni/zeolite-Y

– **Decreased** total methane conversion relative to AC

H_2, CH_4 production and methanation

Ref. Effect of high-pressure pyrolysis on syngas and char structure of petroleum coke, Hyeong-Bin Moon, Chung-Hwan Jeon, *Energy*, 2024



Variations in gas concentration as a function of temperature and pressure

- **H_2** concentration

– Production peaks shift to higher temperatures increasing pressure

– **H_2** decreases due to methanation under high pressure

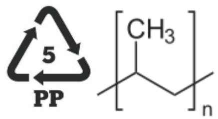
(methanation : $C + 2H_2 \rightarrow CH_4$)

- **CH_4** concentration

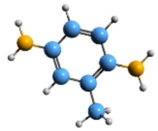
– At 0.1MPa, **CH_4** peaks at 600°C then rapidly declines

– Higher **CH_4** observed above 700°C under high pressure

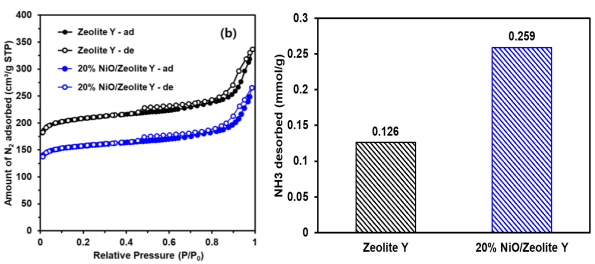
Objective



Waste plastic

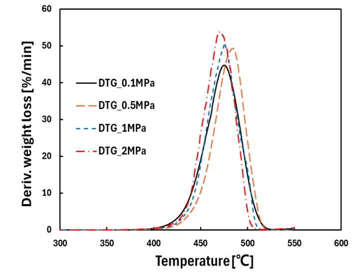


Catalyst

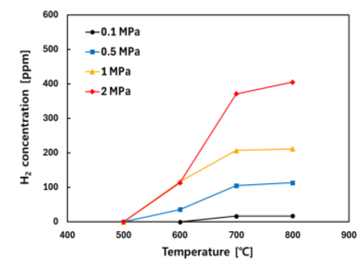


Catalyst characteristics

HP-TGA



Gas chromatography



Pyrolysis

Pyrolysis characteristics under **high-pressure**

Gas analysis

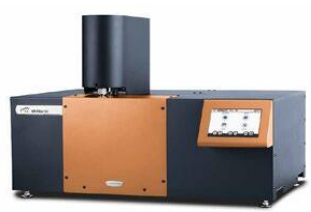
Syngas composition (H_2, CH_4, CO, CO_2)

Optimization H_2

- Catalytic pyrolysis under high pressure & temperature conditions
- Thermogravimetric analysis and gas conversion
- Analysis of catalytic effects on gas concentration
- Maximization **hydrogen production**

Experimental apparatus

Pressurized TGA



HP TGA 75

– High pressure pyrolysis of polypropylene

– Analysis of weight loss under different pressure

– Analysis of the effects of pressure on primary pyrolysis and secondary pyrolysis

▪ **Experimental conditions**

Temperature	10°C/min to 800°C
Pressure	0.1 MPa, 0.5 MPa, 1 Mpa, 2 Mpa
Purge gas	N_2 gas, 30ml/min

Gas chromatography

YL 6500 GC

– Analysis of syngas after high pressure pyrolysis

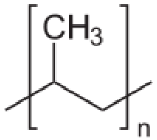
– Using standard gas (H_2, CO, CO_2, CH_4 each 1%, N_2 balance)

– Using TCD and FID

▪ **Experimental conditions**

Temperature	120°C
Measurement Interval	10 min
TCD	H_2
FID	CO, CO_2, CH_4

Sample

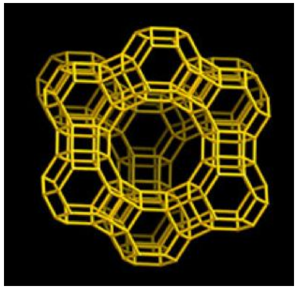


Polypropylene pellet

▪ **Sample conditions**

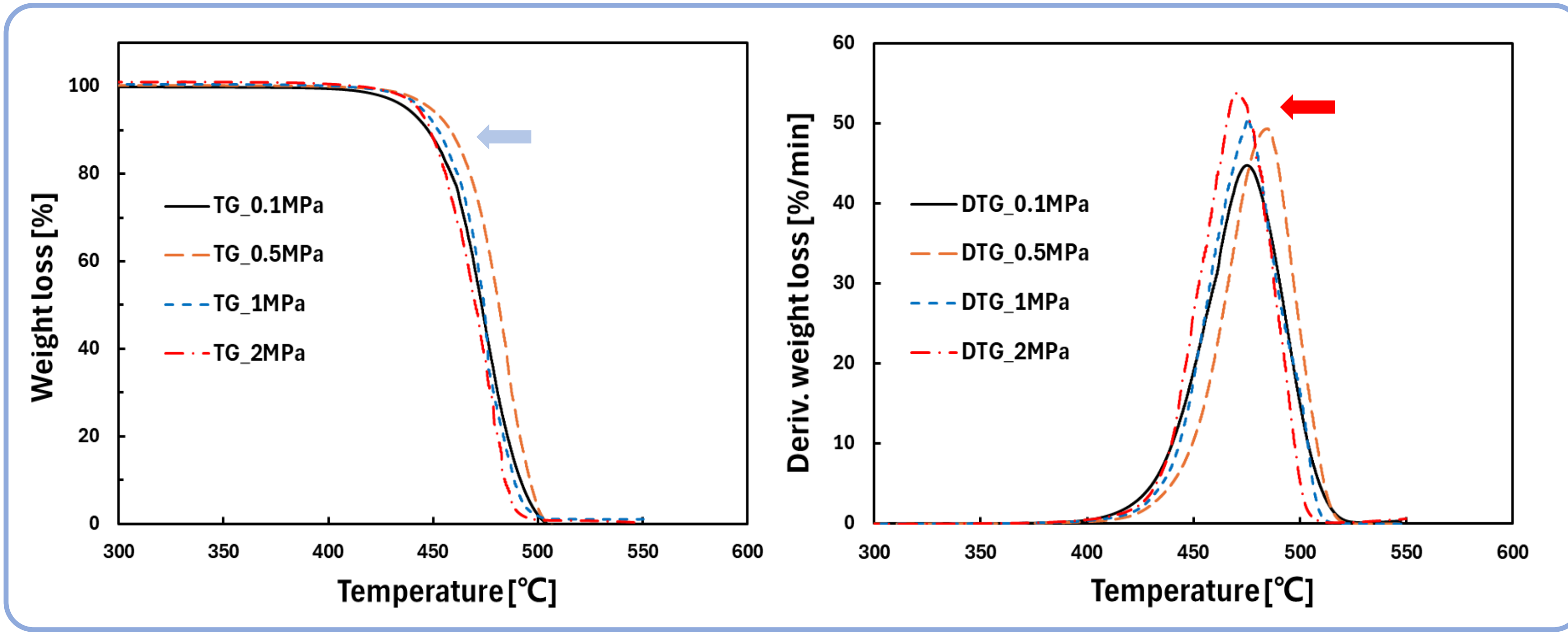
Drying Temperature	40°C for 24h
Weight	25±0.5mg

Catalyst



Zeolite-Y & 20% NiO/Zeolite-Y

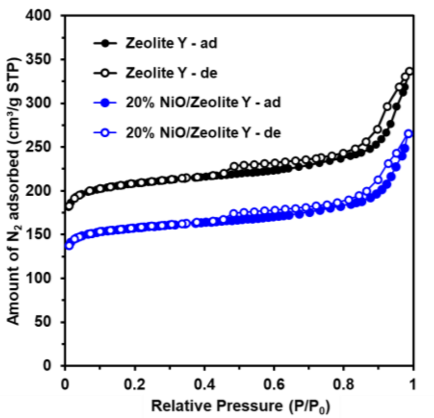
Thermal weight loss decomposition of polypropylene



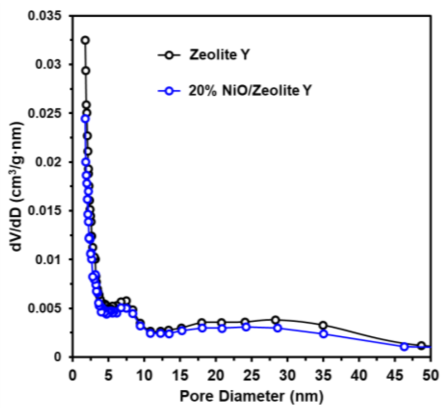
- The onset of pyrolysis was delayed at O.5MPa due to pressure effects [Primary pyrolysis inhibition]
- Confirmation that the trend of the graph shifted to the left as the pressure increased
- As the pressure increases(>0.5MPa), pyrolysis proceeds more rapidly [Promotion of Secondary pyrolysis]

→ **Expected increase in Hydrogen production**

Catalyst characteristics



BET analysis result

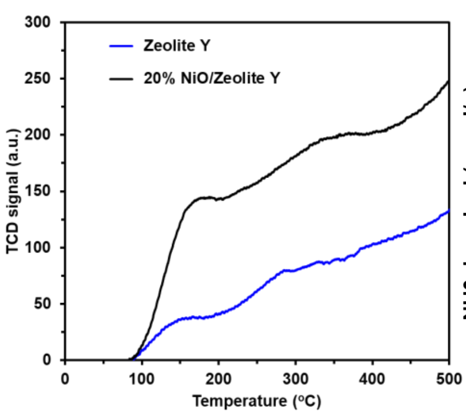


BJH pore-size distribution

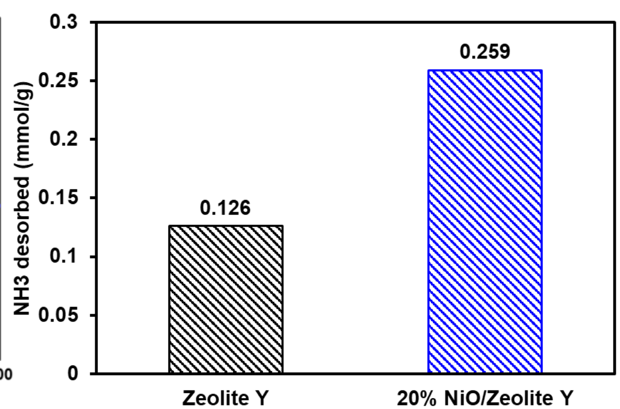
Catalyst	S_{BET} (m^2/g)	V_{total} (cm^3/g)	Pore size (cm^3/g)
Zeolite-Y	704	0.52	1.71
20% NiO/Zeolite-Y	554	0.41	1.70

Slightly lower surface area, but improved catalytic potential:

– Ni is finely dispersed in micropores, promoting active-site accessibility



NH_3-TPD profiles



Total NH_3 desorption amounts

20% NiO/Zeolite-Y exhibits enhanced catalytic features for hydrogen production:

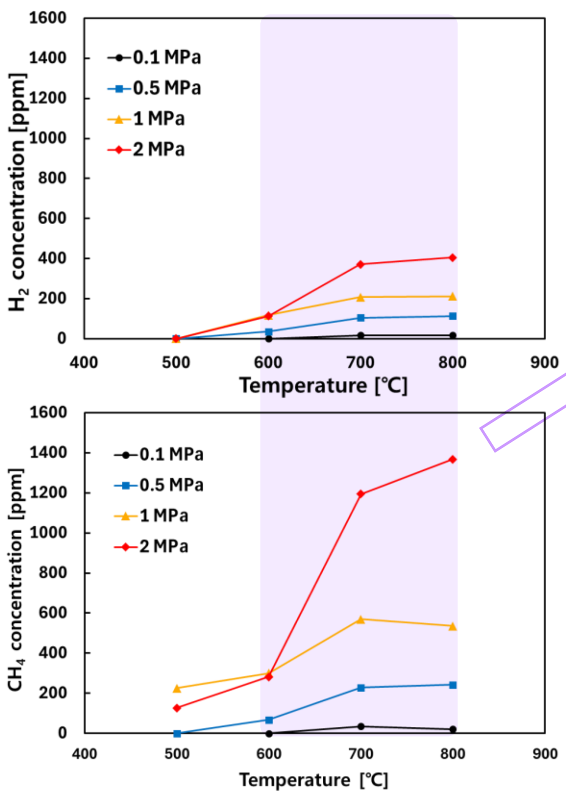
– Stronger acid sites confirmed by increased NH_3 desorption and high-temperature peaks

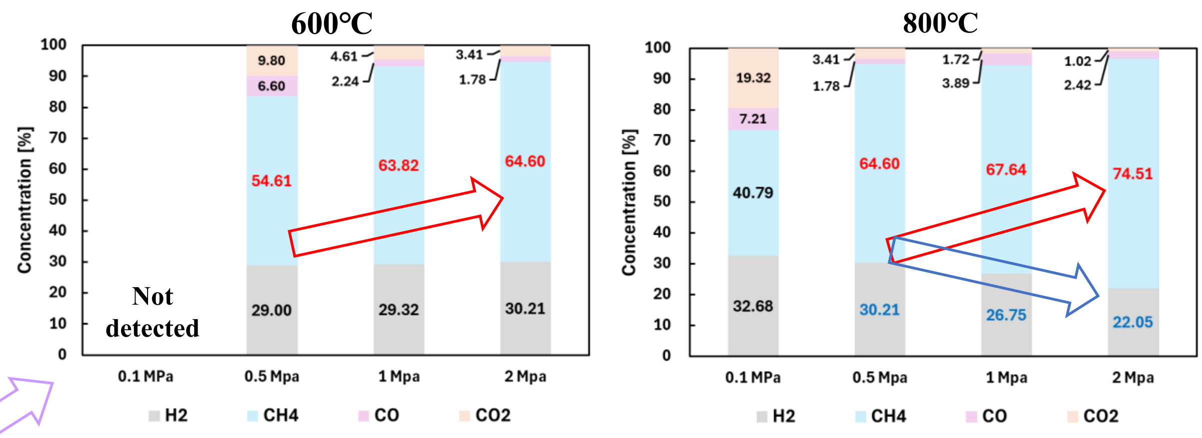
– Improved acid site density despite reduced surface area

– NiO loading promotes active metal-support interaction

Gas composition of PP (Non-catalyst)

Non-catalyst





Gas composition analysis during high-pressure pyrolysis of polypropylene[%]

- As the pressure increases, both H_2 and CH_4 are enhanced

  (Enhanced secondary pyrolysis)

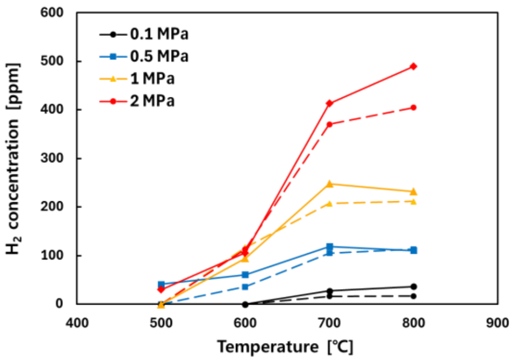
- At early reaction stage (600°C), H_2 maintains while CH_4 increases at high pressure
- At later reaction stage (800°C), H_2 decreases and CH_4 increases at high pressure

  (Activation of methanation reaction)

$$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O \text{ --- (1)}$$
$$CO + 3H_2 \rightarrow CH_4 + H_2O \text{ ------ (2)}$$

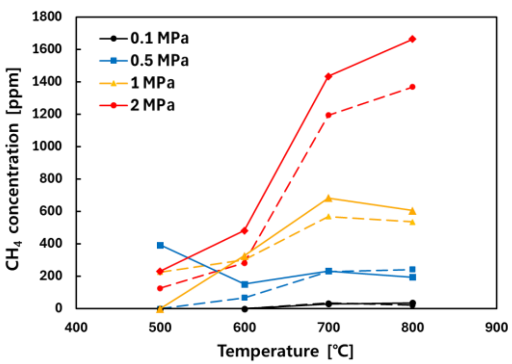
Gas composition of PP (Non-catalyst & Zeolite-Y)

• 점선 : Non-catalyst • 실선 : Zeolite-Y



Non-catalyst				Zeolite-Y			
H_2(ppm)	600℃	700℃	800℃	H_2(ppm)	600℃	700℃	800℃
0.1MPa	0	16.41	17.1	0.1MPa	0	27.43	36.78
0.5MPa	35.96	105.07	113.52	0.5MPa	61.08	119.31	110.78
1MPa	117.13	207.28	211.92	1MPa	94.97	248.21	232.53
2MPa	113.26	370.75	**404.82**	2MPa	106.12	413.36	**489.33**

• 점선 : Non-catalyst • 실선 : Zeolite-Y



Non-catalyst				Zeolite-Y			
CH_4(ppm)	600℃	700℃	800℃	CH_4(ppm)	600℃	700℃	800℃
0.1MPa	0	34.37	21.34	0.1MPa	0	29.45	36.45
0.5MPa	67.72	228.67	242.77	0.5MPa	151.67	233.06	194.48
1MPa	300.49	569.09	535.84	1MPa	325.88	683.29	604.73
2MPa	281.29	1193.97	1368.14	2MPa	481.48	1434.48	1664.05

- Both H_2 and CH_4 **increased** as reaction time and pressure were raised
- Catalytic reaction activated conversion (Oil → Syngas)
- **CH_4** production **increased** under high pressure and high temperature due to **methanation** reaction

$$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O \text{ -- (1)}$$
$$CO + 3H_2 \rightarrow CH_4 + H_2O \text{ ----- (2)}$$

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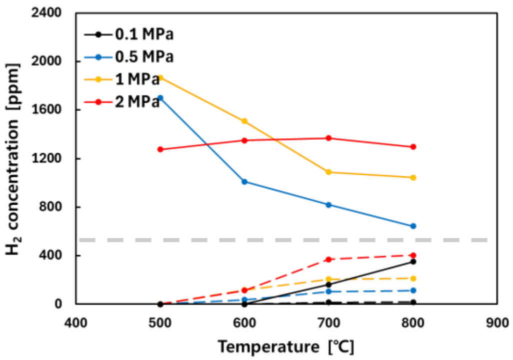
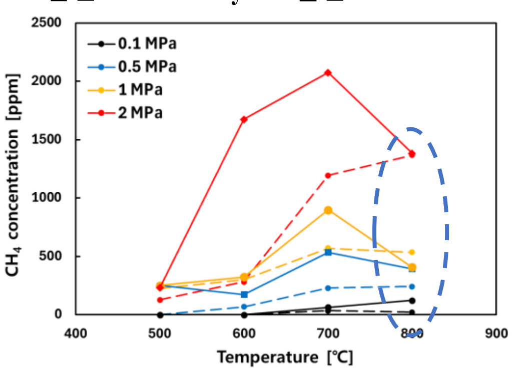
Gas composition of PP (Non-catalyst & NiO/Zeolite-Y)

• 점선 : Non-catalyst • 실선 : NiO/Zeolite-Y



	Non-catalyst				NiO/Zeolite-Y		
H_2(ppm)	600°C	700°C	800°C	H_2(ppm)	600°C	700°C	800°C
0.1MPa	0	16.41	17.1	0.1MPa	0	162.47	350.47
0.5MPa	35.96	105.07	113.52	0.5MPa	1011.10	819.23	642.74
1MPa	117.13	207.28	211.92	1MPa	**1510.08**	1089.49	1045.49
2MPa	113.26	370.75	**404.82**	2MPa	1349.48	1369.13	1297.01

• 점선 : Non-catalyst • 실선 : NiO/Zeolite-Y



	Non-catalyst				NiO/Zeolite-Y		
CH_4(ppm)	600°C	700°C	800°C	H_2(ppm)	600°C	700°C	800°C
0.1MPa	0	34.37	21.34	0.1MPa	0	63.08	123.31
0.5MPa	67.72	228.67	242.77	0.5MPa	172.40	534.61	393.49
1MPa	300.49	569.09	535.84	1MPa	322.63	897.75	406.09
2MPa	281.29	1193.97	1368.14	2MPa	1673.42	2076.03	1383.90

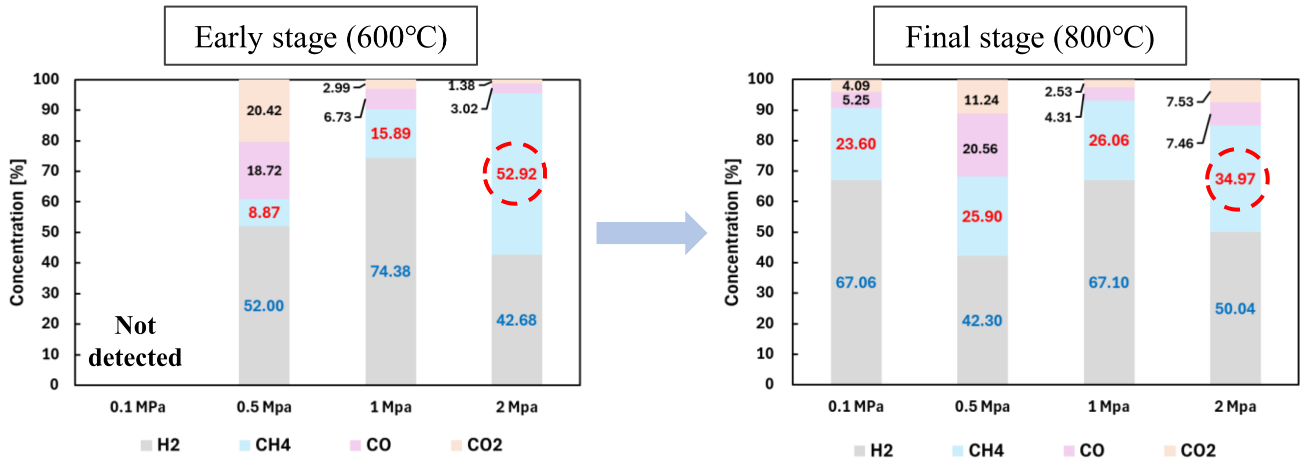
- **H_2** yield significantly **increased** compared to the case without catalyst
- **CH_4** concentration exhibited a **declining trend** at 800°C
- Suppression of the methanation reaction (1) Thermal decomposition of **CH_4**

  (2) Reverse Methanation reaction

Results & Discussion

Gas composition of PP (Non-catalyst & NiO/Zeolite-Y)

NiO/Zeolite-Y



(1) Thermal decomposition of **CH_4** (52.92% → 34.97%)

- **CH_4** decomposed to **H_2**/CO/CO_2 at 800℃ (cracking and reforming reaction)

  → **H_2**, CO, CO_2 concentration increase

(2) Reverse methanation reaction

- The catalyst strongly promotes the dominance of the reverse methanation reaction at high temperature

$$CO_2 + 4H_2 \leftrightarrow CH_4 + 2H_2O \text{ -- (1)}$$

$$CO + 3H_2 \leftrightarrow CH_4 + H_2O \text{ ----- (2)}$$

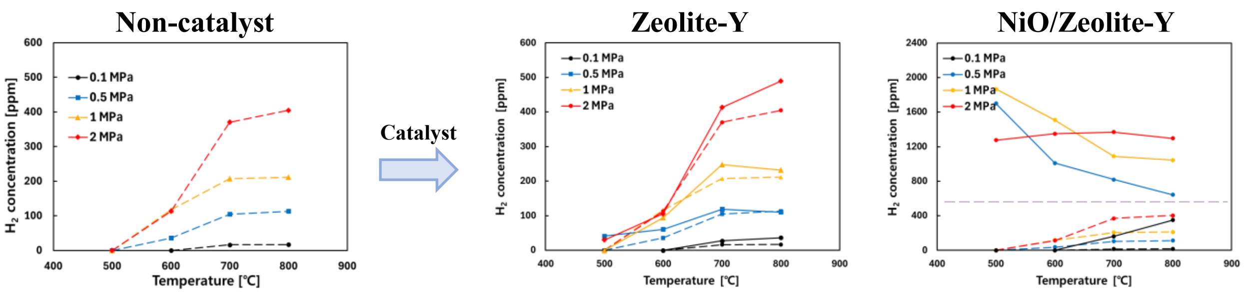
→ CO (3.02% → 7.46%), CO_2 (1.38% → 7.53%)

<u>NiO/Zeolite-Y</u>

- promoting active-site accessibility
- Improved acid site density
- NiO promotes active interaction

Conclusion

- The pyrolysis of polypropylene
  - Secondary pyrolysis was enhanced under high-pressure conditions
  - Increased pyrolysis reactivity results in higher gas yield



- Gas concentration

  (1) Non-catalyst Case
  - As pressure increases, both $\mathbf{H_2}$ and $\mathbf{CH_4}$ increase
  - Under high-temperature & pressure, $\mathbf{H_2}$ decreased while $\mathbf{CH_4}$ increased via the methanation reaction

  (2) Zeolite-Y Case
  - Due to the catalytic reaction, the conversion of oil to gas was enhanced

  (3) NiO/Zeolite-Y Case
  - $\mathbf{H_2}$ yield significantly increased compared to the case without catalyst
  - The reverse methanation reaction becomes dominant at high temperature

The researcher would like to thank the following people who have assisted me

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- Lab member : Hyeong-Bin Moon
- Advisor : Ph.D. Jae-Hun Yang (University of Newcastle)
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Thank you
Any Questions?

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